

CHANGJON, Korea, Democratic People's Republic of

WMO: 470610

Lat: 38.733N

Lon: 128.183E

Elev: 34

StdP: 100.92

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.5	-6.7	-21.2	0.6	-5.1	-18.7	0.7	-4.4	11.5	-1.0	9.4	-1.1	3.4	320	0.369

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.0	31.7	25.6	30.4	24.9	29.1	24.2	27.2	30.1	26.3	28.9	25.4	27.9	2.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.4	22.0	29.2	25.5	20.8	28.1	24.6	19.7	27.1	86.6	30.2	82.2	29.1	78.2	28.1	32.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.6	8.6	7.0	-11.4	33.9	2.8	1.8	-13.5	35.2	-15.1	36.2	-16.7	37.2	-18.7	38.5		
			-12.9	28.6	2.8	1.4	-14.9	29.6	-16.5	30.4	-18.1	31.2	-20.1	32.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.5	-0.4	0.7	5.5	11.8	17.0	20.3	24.0	24.4	20.2	15.3	8.5	2.3	(6)
(7)		DBStd	9.29	3.40	3.28	4.07	4.06	3.84	3.00	3.34	2.66	2.29	2.78	3.97	3.85	(7)
(8)		HDD10.0	1076	323	259	152	26	1	0	0	0	2	73	240		(8)
(9)		HDD18.3	2657	581	492	398	199	72	13	1	0	8	99	294	498	(9)
(10)		CDD10.0	1998	0	0	12	80	218	308	434	446	305	165	30	1	(10)
(11)		CDD18.3	536	0	0	0	3	31	71	177	187	63	4	0	0	(11)
(12)		CDH23.3	3643	0	0	0	18	209	407	1458	1357	189	5	0	0	(12)
(13)		CDH26.7	1027	0	0	0	2	38	92	500	373	21	0	0	0	(13)
(14)	Wind	WSAvg	2.2	2.2	2.2	2.6	3.0	2.6	1.6	1.7	1.6	1.8	2.0	2.3	2.4	(14)
(15)	Precipitation	PrecAvg	1601	43	54	46	75	104	155	327	317	213	103	103	51	(15)
(16)		PrecMax	2360	126	206	102	155	329	459	606	665	542	344	726	257	(16)
(17)		PrecMin	1111	3	1	5	10	17	39	80	102	26	5	10	1	(17)
(18)		PrecStd	295	35	50	32	44	70	116	141	149	135	90	136	64	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.8	12.2	20.2	25.8	29.6	31.3	33.1	33.2	28.7	24.1	20.1	13.4	(19)
(20)		MCWB	5.1	6.3	12.3	15.9	17.7	21.8	25.6	27.4	23.0	17.5	13.9	9.1		(20)
(21)		2%	DB	7.4	9.4	16.5	23.0	27.5	29.1	31.8	31.6	27.0	22.2	17.8	11.3	(21)
(22)		MCWB	3.4	4.9	9.5	13.9	17.0	21.2	25.6	26.6	21.8	17.1	12.6	7.6		(22)
(23)		5%	DB	6.0	7.5	14.5	20.9	25.7	27.3	30.8	30.1	25.3	21.0	16.2	9.2	(23)
(24)		MCWB	2.6	3.5	8.6	12.8	16.1	20.6	25.3	25.5	21.1	16.8	11.8	6.0		(24)
(25)		10%	DB	4.5	5.8	12.2	18.9	23.7	25.8	29.6	28.7	24.1	19.9	14.5	7.7	(25)
(26)		MCWB	1.2	2.3	7.2	11.9	15.4	20.3	24.9	25.2	20.7	16.1	10.9	4.6		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.6	7.1	13.3	18.0	20.8	24.6	28.1	29.0	24.7	20.1	15.6	10.2	(27)
(28)		MDCB	8.2	11.0	18.4	23.1	26.9	28.4	30.9	31.2	27.3	21.8	17.7	12.3		(28)
(29)		2%	WB	4.2	5.3	10.6	15.6	19.1	23.2	27.0	27.7	23.2	18.8	13.7	8.1	(29)
(30)		MDCB	6.8	8.7	15.3	20.6	24.6	26.2	30.3	30.4	25.3	20.8	16.2	10.7		(30)
(31)		5%	WB	2.8	4.1	8.9	14.0	17.9	22.2	26.2	26.7	22.3	17.8	12.6	6.5	(31)
(32)		MDCB	5.3	6.9	13.4	19.0	22.8	25.3	29.4	29.2	24.2	20.0	15.2	8.9		(32)
(33)		10%	WB	1.7	2.9	7.5	12.6	16.8	21.3	25.4	25.8	21.4	16.9	11.4	5.0	(33)
(34)		MDCB	3.8	5.2	11.6	17.5	21.3	24.3	28.4	27.8	23.3	19.0	14.0	7.3		(34)
(35)	Mean Daily Temperature Range	MDBR	6.3	6.1	6.5	7.6	7.6	6.0	4.9	5.0	6.1	6.9	6.5	6.4	(35)	
(36)		5% DB	MCDBR	7.5	8.2	9.7	11.2	11.1	9.7	7.1	6.9	7.8	8.3	8.3	7.5	(36)
(37)		MCWBR	5.7	5.9	6.2	6.2	5.6	5.2	4.1	4.1	5.1	5.6	5.8	6.0	(37)	
(38)		5% WB	MCDBR	6.6	7.6	9.0	10.2	9.8	7.7	6.5	6.0	6.3	7.2	7.0	7.1	(38)
(39)		MCWBR	5.4	5.8	6.3	6.6	6.0	5.0	4.3	4.4	4.8	5.4	5.4	6.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.394	0.453	0.532	0.592	0.599	0.648	0.615	0.537	0.475	0.476	0.440	0.390	(40)	
(41)		taud	2.032	1.903	1.784	1.707	1.744	1.706	1.843	2.047	2.144	2.004	2.024	2.076	(41)	
(42)		Ebn at Noon	735	739	721	708	715	680	699	746	769	710	678	706	(42)	
(43)		Edh at Noon	124	162	202	231	227	236	204	163	139	145	125	111	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.43	3.25	4.14	4.95	5.44	5.16	4.27	4.26	3.99	3.40	2.37	2.15	(44)	
(45)		RadStd	0.22	0.30	0.39	0.42	0.56	0.45	0.58	0.50	0.46	0.29	0.33	0.21	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+104	+55

Nomenclature: See separate page